

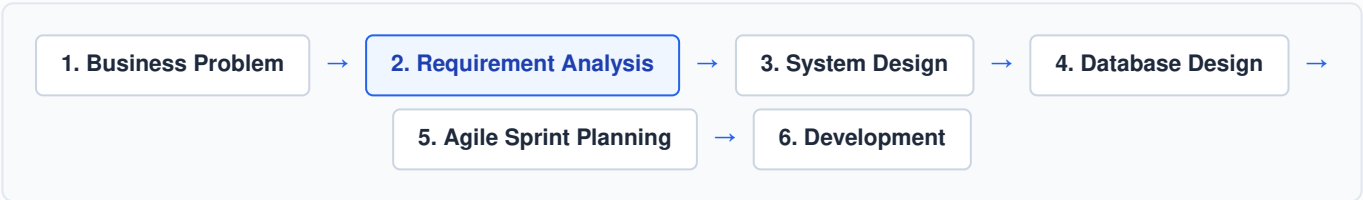
SkillMatch AI

AI-Powered Resume & Job Hiring Platform — Complete Requirement Analysis

Document Version: 1.0.0	Lifecycle Phase: Requirement Analysis	Target Audience: Product & Engineering	Status: Approved / Ready for Design
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01 Position in Project Lifecycle

In software engineering methodology, system building follows a strict sequential progression to ensure business alignment, clarity, and architectural integrity. The placement of **Requirement Analysis** within this lifecycle is foundational:



For **SkillMatch AI**, Requirement Analysis serves as the critical bridge between recognizing hiring inefficiencies and implementing technical solutions. It strictly defines *WHAT* the platform must accomplish before deciding on tech stacks, database schemas, or algorithmic implementation details.

02 What is the Requirement?

Project Name: SkillMatch AI — AI-Powered Resume & Job Hiring Platform

Core Objective: Provide an intelligent, automated platform that standardizes and streamlines the recruitment match lifecycle through NLP and Machine Learning.

System Functional Capabilities Baseline

- **Candidate Resume Upload:** Multi-format resume parsing and feature extraction.
- **Recruiter JD Upload:** Job Description parsing and structural breakdown of requirements.
- **Automated Extraction:** Identification of core competencies, skills, experience, and education.
- **Matching Engine:** Vector/semantic comparison of candidate profiles against job demands.
- **Job Discovery:** Intelligent job recommendations tailored to candidate skill sets.
- **Candidate Ranking:** Dynamic scoring and ranking of applicant pools for recruiters.
- **Explainable Insights:** Transparent breakdowns of match scores, strengths, and identified skill gaps.

03 Business Problem Definition

Problem Statement

The contemporary recruitment ecosystem suffers from acute friction on both sides of the hiring pipeline:

- **Candidates** blindly submit applications without understanding how their skill sets align with specific job criteria, leading to high rejection rates and low engagement.
- **Recruiters & HR Professionals** spend countless hours manually reviewing unstructured resumes. Comparing technical skills, years of experience, educational context, and project depth across hundreds of candidates is error-prone, subjective, and unsustainable.

Core Engineering Problem

"How can we automatically convert unstructured resumes and job descriptions into an understandable, objective, and actionable candidate–job match score accompanied by contextual explanation?"

04 Business Goal & Objectives

To deploy an AI-driven hiring platform that automates candidate-job matching while guaranteeing transparency and explainability for both candidates and recruiters.

Target Persona	Core Business Question	System Solution / Value Delivered
Candidate	"Which jobs am I truly qualified for, and what skills am I missing?"	Targeted job recommendations and clear skill-gap breakdown for career growth.
HR / Recruiter	"Which candidates are most suitable for this role right now?"	Automated resume ranking, rapid screening, and candidate profile synthesis.

05 Stakeholder Analysis

Stakeholders represent individuals or groups interested in or impacted by the system's outcome.

Stakeholder Category	Primary Interests & Motivations
Candidates	Seeking suitable career opportunities, rapid application feedback, and actionable insights on missing skills.
HR / Recruiters	Reducing time-to-hire, minimizing manual resume screening effort, and improving candidate match quality.
Hiring Managers	Receiving high-caliber, pre-qualified candidates who meet technical and experiential job prerequisites.
Administrators	

Stakeholder Category	Primary Interests & Motivations
	Maintaining user account security, platform integrity, system health, and managing system parameters.
Company Management	Maximizing operational hiring efficiency, reducing recruitment overhead costs, and scalable talent acquisition.

06 Actor Profiles & Capabilities

Actors represent direct human or automated entities interacting directly with the system interface or APIs.

Actor Role	System Capabilities & Interaction Scope
Candidate	Register/Login • Upload Resume (PDF/DOCX) • View Extracted Skill Profile • View Job Recommendations • View Detailed Match Scores & Skill Gaps • Submit Job Applications.
HR / Recruiter	Register/Login • Create & Manage Job Postings • Upload Job Descriptions • View Extracted Job Requirements • View Ranked Applicant List • Side-by-side Candidate Comparison • View Match Insights • Shortlist / Reject Candidates.
Administrator	Manage User Accounts & Permissions • Approve/Manage Corporate Recruiter Accounts • Manage Job Categories & Taxonomy • Monitor System Telemetry & Logs • Manage Reported Content • Generate System Analytics Reports.

07 Functional Requirements (FR)

Functional requirements define specific behaviors and capabilities that the system **MUST** provide.

Candidate Domain

Req ID	Requirement Description
FR-01	The system shall allow candidates to create a user account with valid credentials.
FR-02	The system shall authenticate candidate login requests securely.
FR-03	The system shall allow candidates to upload resume files in PDF and DOCX formats.
FR-04	The system shall extract structured text and semantic features from uploaded resumes.
FR-05	The system shall identify technical and soft skills from the parsed resume data.
FR-06	The candidate shall be able to view and edit extracted skills on their profile.
FR-07	The candidate shall be able to view personalized job recommendations.
FR-08	The candidate shall be able to view a percentage match score for any given job posting.
FR-09	The candidate shall be able to view matched skills and missing skill gaps for a job posting.

Req ID	Requirement Description
FR-10	The candidate shall be able to submit a direct application for open positions.

HR / Recruiter Domain

Req ID	Requirement Description
FR-11	The system shall allow HR recruiters to register corporate accounts.
FR-12	The system shall allow HR recruiters to create job postings and upload Job Descriptions (JDs).
FR-13	The system shall automatically extract mandatory and preferred skill requirements from uploaded JDs.
FR-14	The system shall display a list of matching candidates for each active job posting.
FR-15	The system shall rank candidate profiles automatically based on multi-factor role fit algorithms.
FR-16	The HR recruiter shall be able to compare candidate profiles side-by-side.
FR-17	The system shall display detailed candidate strengths and missing skill gaps to the recruiter.
FR-18	The HR recruiter shall be able to update applicant status (Shortlist, In Review, Rejected).

Administrator Domain

Req ID	Requirement Description
FR-19	The administrator shall be able to manage all registered user accounts (suspend, activate, delete).
FR-20	The administrator shall be able to review and approve recruiter corporate credentials.
FR-21	The administrator shall be able to create, edit, and delete job categories and skill taxonomies.
FR-22	The administrator shall be able to monitor system usage metrics and active sessions.
FR-23	The administrator shall be able to generate and download system usage and performance reports.

08 Non-Functional Requirements (NFR)

Non-Functional Requirements establish system quality attributes, operational constraints, and performance benchmarks.

Attribute	NFR ID	Specification Standard
Security	NFR-01	RBAC enforcement; candidate resumes and PII must be encrypted at rest (AES-256) and in transit (TLS 1.3). Only authorized recruiters may access applicant details.
Performance	NFR-02	

Attribute	NFR ID	Specification Standard
		Resume parsing and skill extraction must complete within < 5 seconds. Matching calculations for search queries must execute in < 2 seconds.
Scalability	NFR-03	System architecture must scale horizontally to support 100,000+ active candidates, 10,000+ concurrent job postings, and high-frequency matching operations.
Reliability	NFR-04	Parsing accuracy for standard resume formats must maintain a minimum 95% success rate without runtime engine crashes.
Availability	NFR-05	The platform must maintain 99.9% uptime operational availability, excluding scheduled off-peak maintenance windows.
Explainability	NFR-06	Every generated match score must be accompanied by explicit textual justification and breakdown metrics (e.g., Skill Match % + Experience Fit %).
Usability	NFR-07	UI designs must adhere to WCAG 2.1 Level AA accessibility guidelines, ensuring intuitiveness for non-technical candidates and recruiters.

09 Functional vs. Non-Functional Matrix

Functional Requirements (WHAT)	Non-Functional Requirements (HOW WELL)
Candidate uploads resume file FR	Resume parsing completes within 5 seconds securely NFR
Recruiter creates job posting FR	Job system handles 10,000+ active postings seamlessly NFR
System calculates candidate match score FR	Match results include human-understandable explanation NFR
Recruiter shortlists candidates FR	System guarantees 99.9% platform availability NFR

10 Requirement Classification Examples

Operational classification reference for specification verification:

- "Candidate should be able to upload a resume." → **FUNCTIONAL REQUIREMENT**
- "The resume should only be accessible to authorized users." → **NON-FUNCTIONAL REQUIREMENT (SECURITY)**
- "HR should be able to shortlist a candidate." → **FUNCTIONAL REQUIREMENT**
- "The application should respond quickly to matching requests." → **NON-FUNCTIONAL REQUIREMENT (PERFORMANCE)**
- "The system should calculate a candidate's match score." → **FUNCTIONAL REQUIREMENT**
- "The platform should support a large number of candidates." → **NON-FUNCTIONAL REQUIREMENT (SCALABILITY)**

11 Agile User Stories

User stories capture capability requirements structured from user persona perspectives: As a [user], I want [functionality], so that [benefit].

- **US-01 (Resume Upload):** As a candidate, I want to upload my resume so that the system can automatically extract my skills and recommend relevant job opportunities.
- **US-02 (Job Description Parsing):** As an HR recruiter, I want to upload or paste a job description so that the platform can isolate key technical requirements and identify qualified talent.
- **US-03 (Skill Gap Visibility):** As a candidate, I want to view my missing skills for a target job so that I can understand how to upskill effectively.
- **US-04 (Automated Candidate Ranking):** As an HR recruiter, I want candidates ranked by job compatibility so that I can focus my manual screening on top-tier applicants.
- **US-05 (Match Explainability):** As an HR recruiter, I want transparent feedback explaining why a candidate scored a specific match percentage so that I can make unbiased hiring decisions.
- **US-06 (User Account Governance):** As an administrator, I want to manage platform user accounts so that system access remains secure and compliant.

12 Acceptance Criteria Specifications

Defines exact validation parameters required to mark a user story as "Done".

User Story: US-01 (Candidate Resume Upload & Extraction)

Acceptance Criteria Checklist:

- ✓ Candidate must be logged into an active candidate session.
- ✓ Candidate can upload files in .pdf or .docx formats up to 5MB.
- ✓ System processes file through parsing pipeline without error.
- ✓ Candidate contact, work history, and education fields are parsed into structured data.
- ✓ System identifies and highlights key technical skills.
- ✓ Extracted profile data is safely persisted to candidate database record.
- ✓ UI displays extracted profile to candidate for verification/editing within 5 seconds.

13 Behavior-Driven Scenarios (Given–When–Then)

Scenario 1: Candidate Resume Processing

GIVEN the candidate is logged into the platform and possesses a valid PDF resume
WHEN the candidate uploads their resume to the profile section
THEN the system should extract skill tags and update the candidate profile record automatically.

Scenario 2: Automated Job Matching & Gap Analysis

GIVEN an HR recruiter has published a valid job description with target skill requirements
WHEN the automated matching engine processes registered candidate profiles
THEN the system should compute match scores and render a list of matched vs missing skills for each candidate.

Scenario 3: HR Shortlisting Action

GIVEN an HR recruiter is viewing the ranked candidate list for an open posting
WHEN the recruiter clicks the "Shortlist Candidate" action button
THEN the candidate status should update to "Shortlisted" and trigger an notification to the candidate.

14 Requirement-to-Development Traceability

The system establishes a strict, traceable lineage from business intent down to sprint execution tasks:

- 1. **Business Requirement:** Recruiter requires efficient candidate screening mechanism.
 - ↓ 2. **User Story (US-02):** As HR, I want to upload a JD so that I can match candidates.
 - ↓ 3. **Acceptance Criteria:** Extraction of required skills, match scoring, rank ordering.
 - ↓ 4. **System Design:** HR Frontend UI → REST API → AI Parser & Engine → Postgres DB.
 - ↓ 5. **Database Schema Entities:** User, Candidate, Resume, Skill, Job, MatchResult, Application.
 - ↓ 6. **Development Task:** Build JD Upload REST Endpoint & Integration with NLP Parser.
 - ↓ 7. **Sprint Allocation:** Sprint 2 (Core Processing Engine Epic).

15 Requirement Validation Checklist

Validation Category	Core Assessment Questions	SkillMatch AI Project Verification
1. Business Alignment	What problem is solved? What is the core business objective? Who are stakeholders?	✓ Solves manual resume screening lag; automates matching with explainable scores.
2. User Roles	Who uses the system? What can each role accomplish?	✓ Candidate, Recruiter, and Admin roles fully mapped with clear permission scopes.
3. Functional Coverage	What explicit features must the system implement?	✓ 23 distinct Functional Requirements specified across all domains.
4. Non-Functional Quality	How must the system operate under constraints?	✓ Explicit SLAs set for Security, Speed (<2s), Scalability, and Availability (99.9%).

Validation Category	Core Assessment Questions	SkillMatch AI Project Verification
5. Quality Criteria	Are requirements Clear, Complete, Testable, and backed by Acceptance Criteria?	✓ Verified via User Stories, Acceptance Criteria, Given-When-Then, and Traceability Matrix.

Next Architectural Stage

With the Requirement Analysis phase officially complete and validated, the SkillMatch AI project is fully prepared to proceed to **Phase 3: System Design** (Architecture, API Endpoints, and Component Interfaces) followed by **Phase 4: Database Design**.